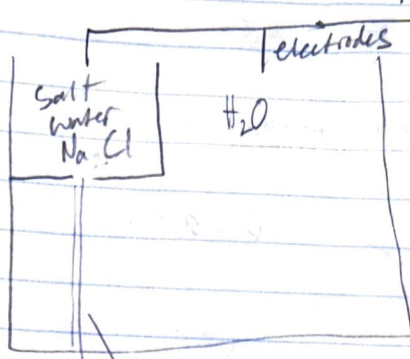


Lecture 09/30: Flario



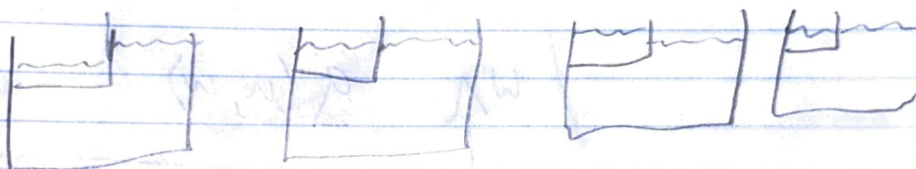
⇒ will read signal of density gradient of salt water

[1st - order approximation of an animal cell]

thin inlet ⇒ laminar flow to bottom, little mixing

→ closed systems don't oscillate indefinitely, need energy input

→ this reaction will eventually 'settle' and the oscillation of the height of fluid



periodic

→ Limit cycle: a trajectory in phase space that the system returns to when perturbed

→ this system is an example of a Limit Cycle

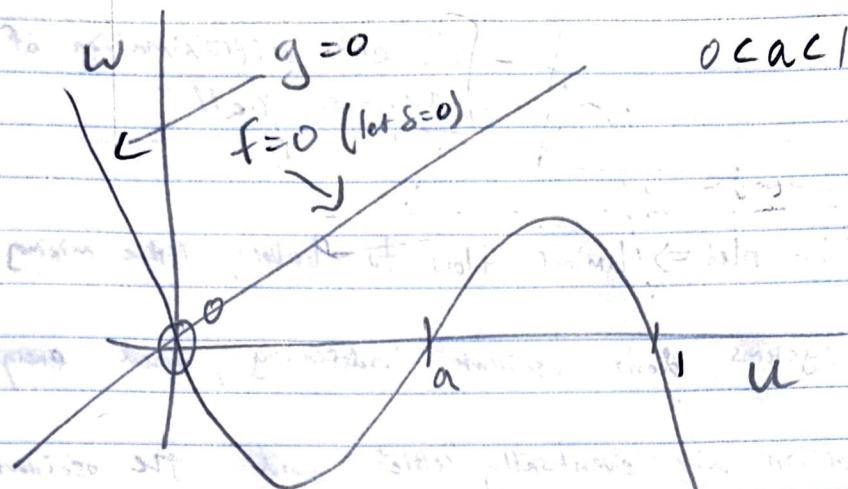
→ the system is

$$\frac{\partial u}{\partial t} = g(u, w) = w(1-u)(a-u) - w$$

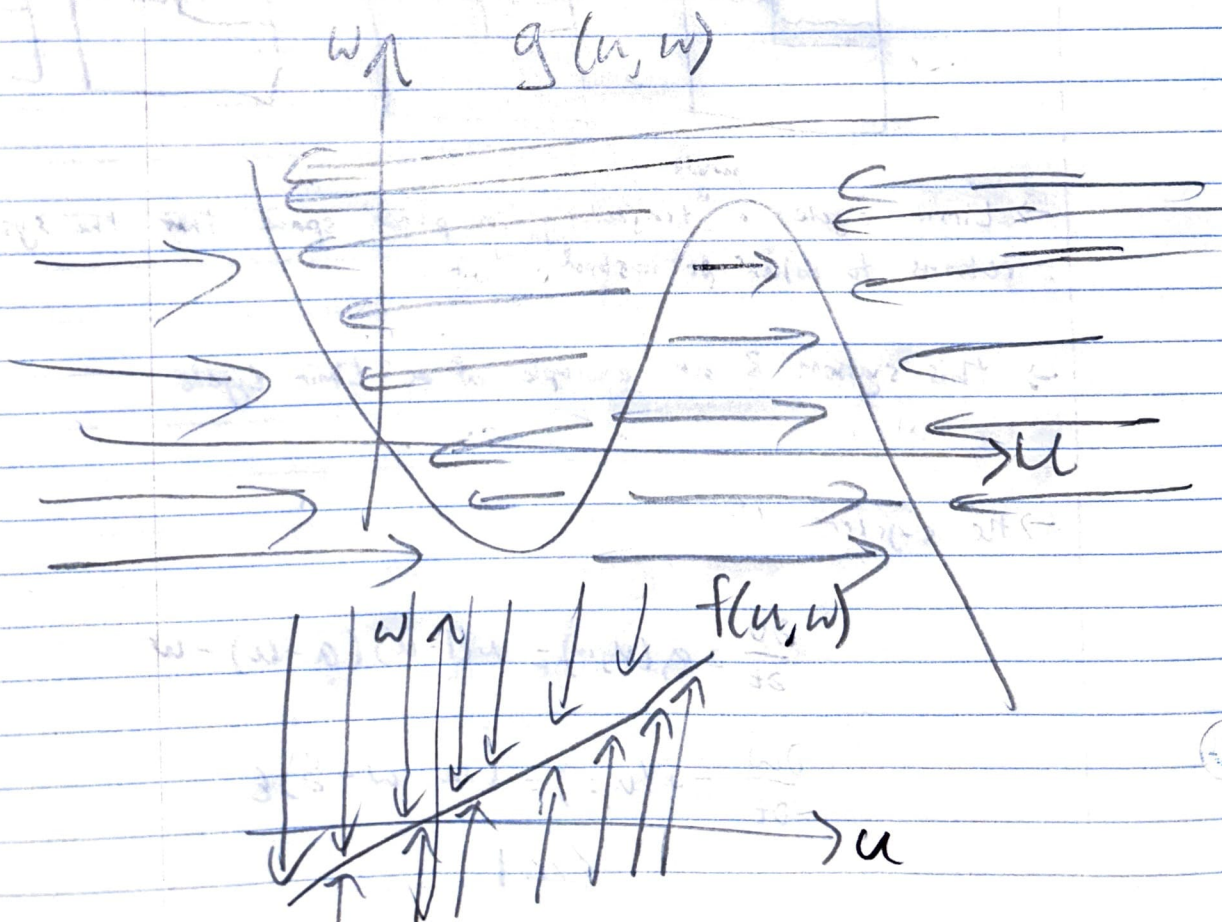
$$\frac{\partial w}{\partial t} = f(u, w) = (u - w - \delta)\epsilon$$

ε < 1

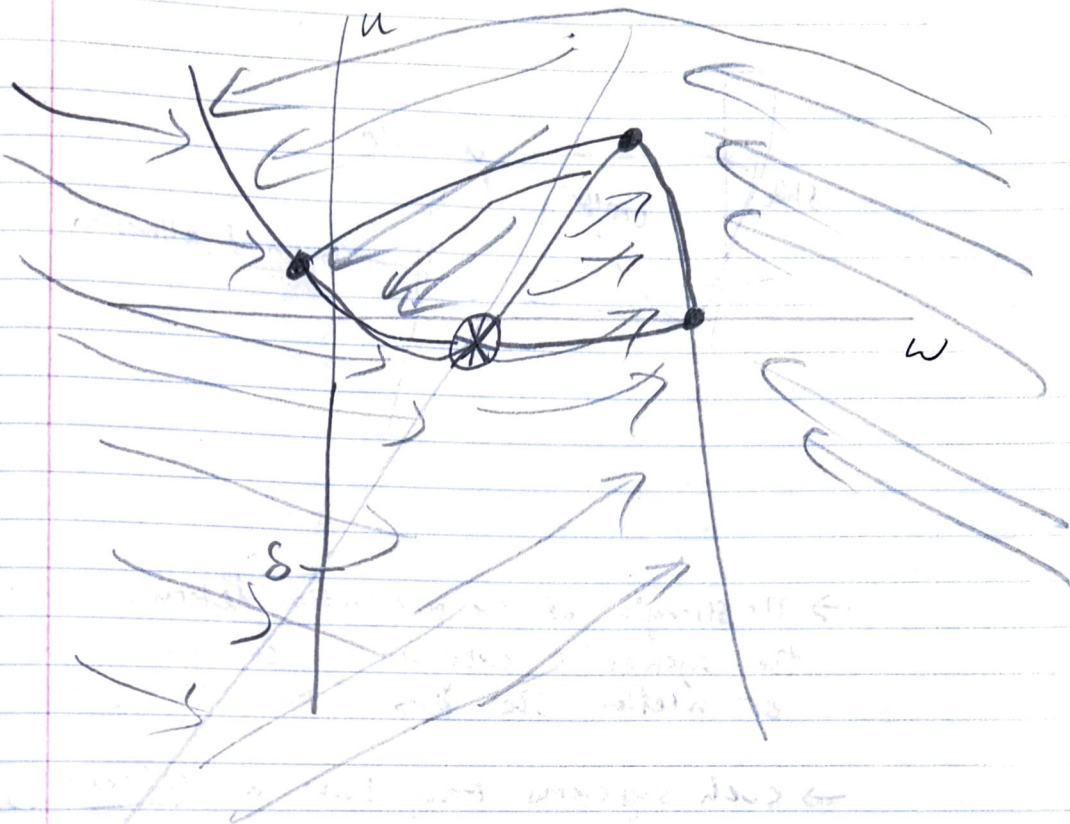
→ consider the system nullclines:



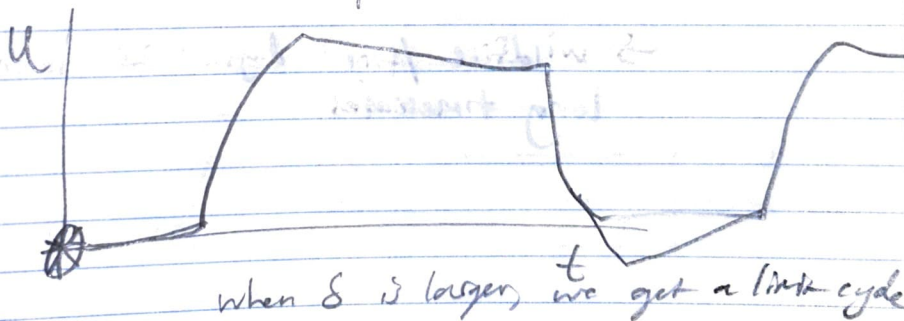
⇒ fixed point @ the origin



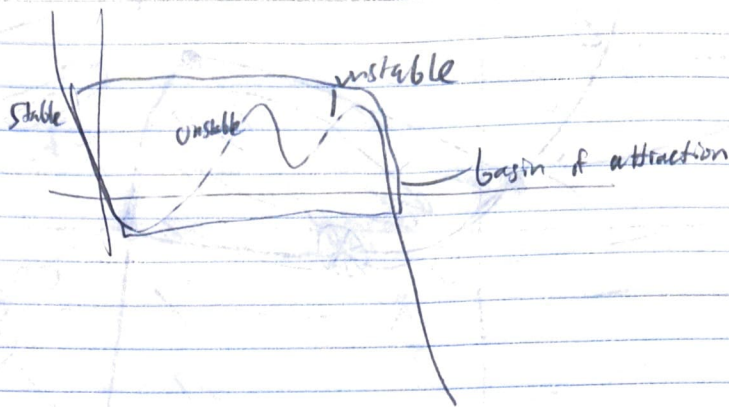
The superposition of the fluxes for each coordinate:



Closed orbits!
action potential



Now, consider $g \in \mathcal{P}^{(5)}(u)$



→ The strength of the perturbation determines whether the system is sufficiently stimulated for activation or whether the excitation returns to stable equilibrium

→ such systems may have a refractory period

→ an example is a toilet, where the initial perturbation is the flush

→ wildfire / fire dynamics follow this on long timescales